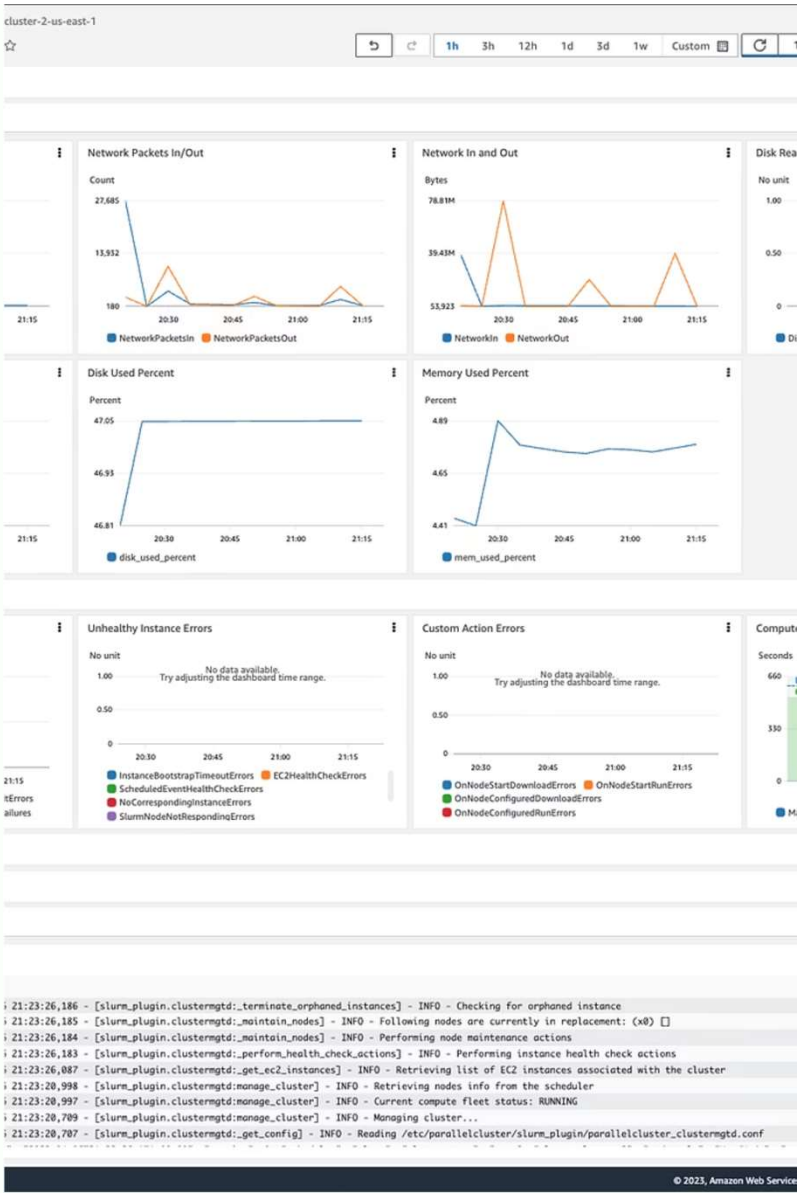
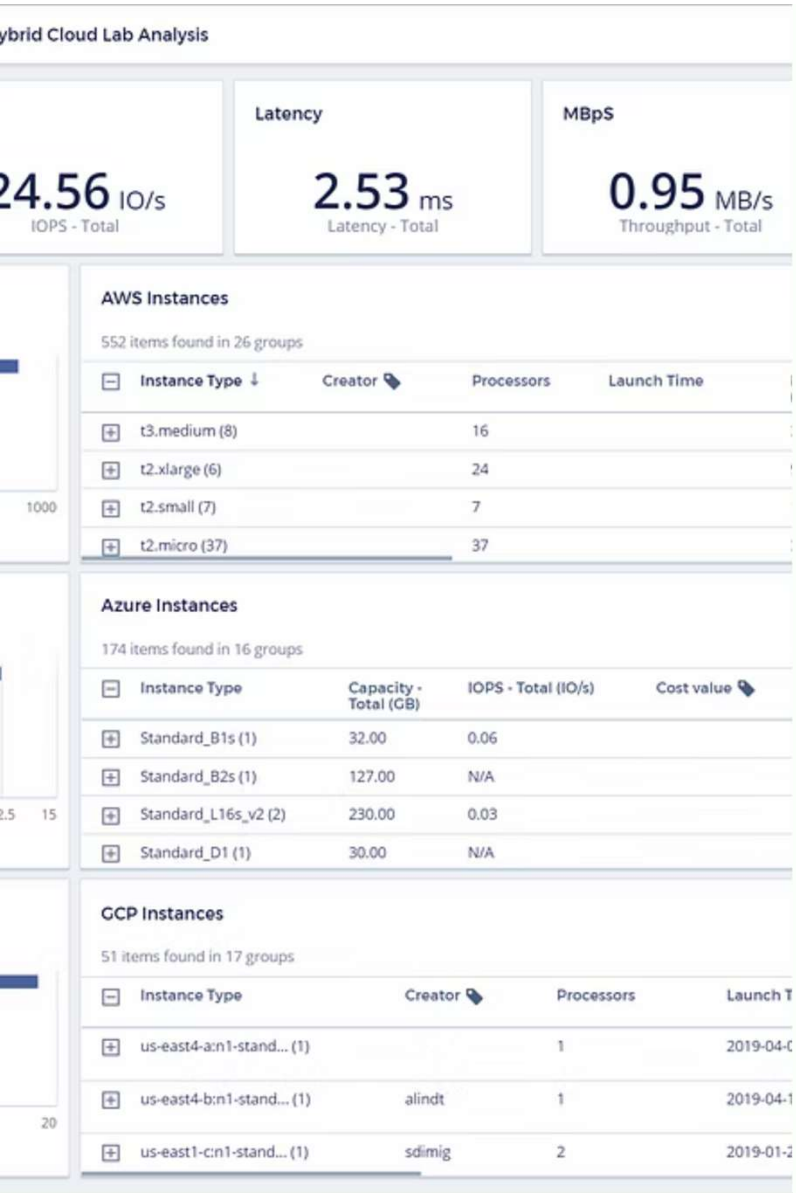
Monitoring & Logging in AWS

CloudWatch, CloudTrail, AWS Config, and Trusted Advisor

A comprehensive guide to understanding AWS's powerful monitoring, logging, and compliance tools to optimize your cloud infrastructure.





Session Objectives

AWS Monitoring Tools

Understand the ecosystem of AWS monitoring & logging tools and how they complement each other

CloudWatch Mastery

Learn CloudWatch's key features including Logs, Metrics, and Alarms for comprehensive resource monitoring

Auditing & Compliance

Explore CloudTrail auditing capabilities and AWS Config compliance tracking mechanisms

Best Practices

Understand how Trusted Advisor identifies optimization opportunities across your AWS environment

Part 1: Amazon CloudWatch

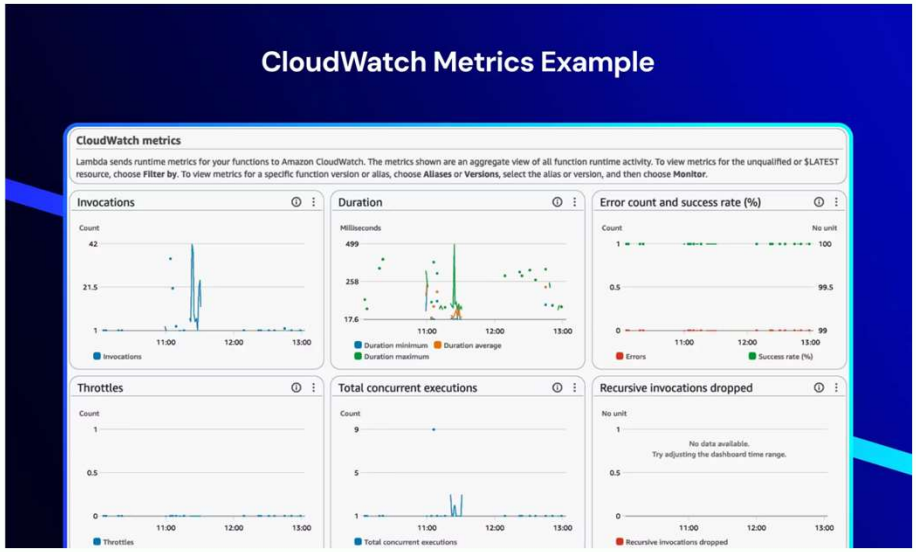
Real-time monitoring for AWS resources and applications

CloudWatch Overview

Amazon CloudWatch provides real-time monitoring of your AWS resources and applications. It collects and tracks metrics, monitors log files, and responds to changes in your AWS environment.

Key Components

- Metrics: Time-ordered data points
- Logs: Aggregation and analysis
- Alarms: Threshold-based notifications
- Dashboards: Custom visualization



Data Sources: AWS Services, Custom metrics, CloudWatch Agent

CloudWatch Metrics

CloudWatch Metrics are time-ordered data points that represent the performance of your AWS resources and applications over time.

EC2 Metrics

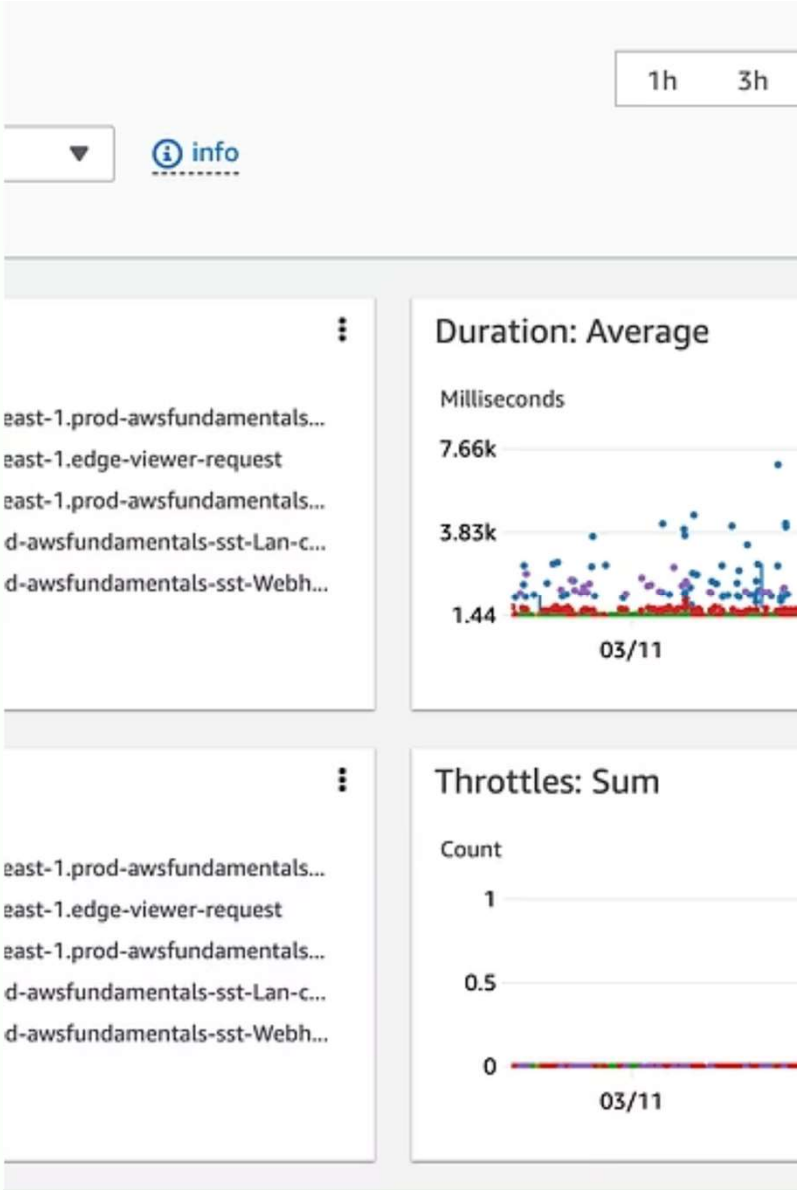
- CPUUtilization
- NetworkIn/NetworkOut
- DiskReadOps/DiskWriteOps
- StatusCheckFailed

RDS Metrics

- FreeStorageSpace
- DatabaseConnections
- ReadIOPS/WriteIOPS
- CPUUtilization

Custom Metrics

Use the put-metric-data API to publish your own metrics from applications or services



CloudWatch Logs

CloudWatch Logs enables you to centralize logs from all your systems, applications, and AWS services for monitoring, storage, and analysis.

Sources:

- CloudWatch Agent (EC2, on-premises)
- AWS Lambda functions
- API Gateway
- Amazon ECS containers
- Many other AWS services

Key Features:

Log Groups & Streams

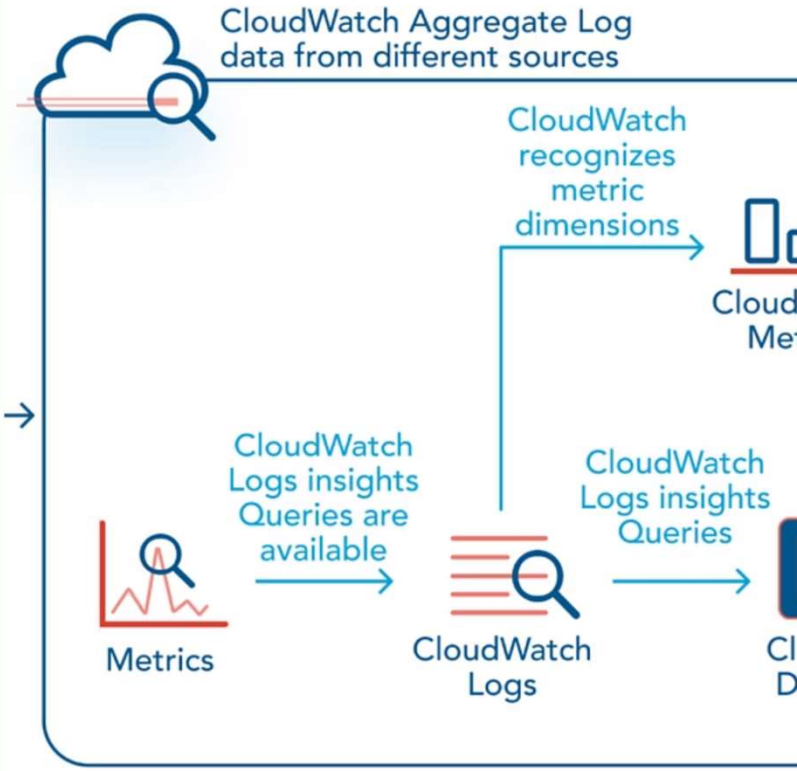
Organize logs by application and instance

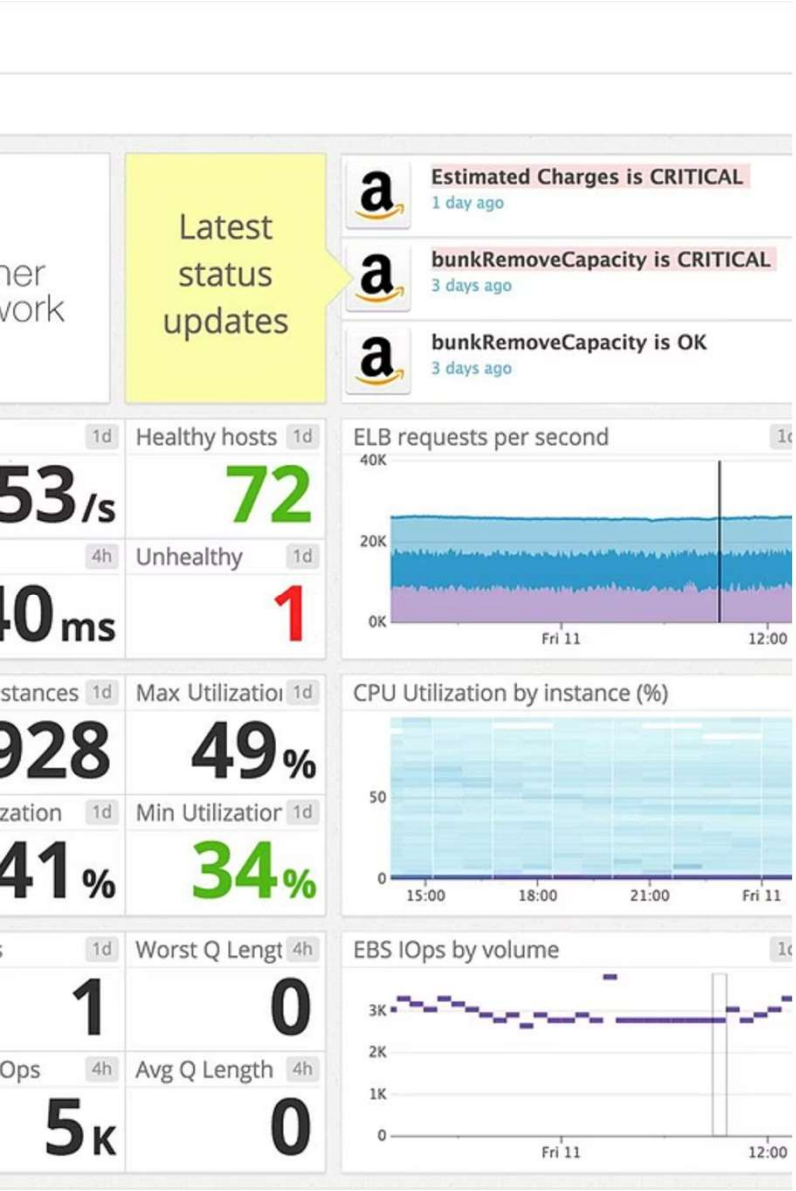
Metric Filters

Extract metrics from log data

Retention Policies

Control storage duration and costs





CloudWatch Alarms & Dashboards

CloudWatch Alarms

Alarms watch a single metric over a specified time period and perform actions based on the metric's value relative to a threshold.

Example: CPU > 80% for 5 minutes → Send SNS notification → Auto Scaling or remediation

- State change: OK, ALARM, INSUFFICIENT_DATA
- Evaluation periods: How many periods to evaluate
- Datapoints to alarm: How many datapoints must breach

CloudWatch Dashboards

Customizable home pages in the CloudWatch console that you can use to monitor your resources in a single view.

- Cross-region visibility
- Multiple metrics on one graph
- Shareable with team members
- Programmatic creation via API

Part 2: AWS CloudTrail

Comprehensive API auditing and governance

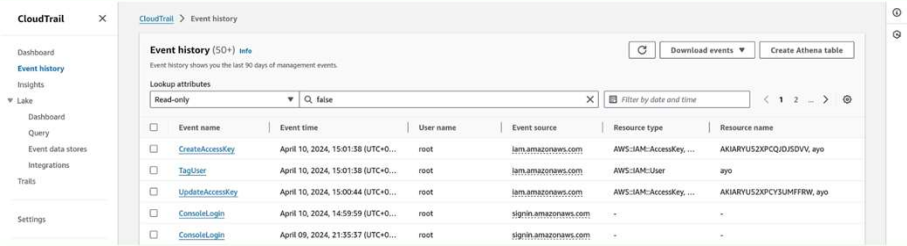
CloudTrail Overview

AWS CloudTrail is a service that enables governance, compliance, operational auditing, and risk auditing of your AWS account. It records and logs all API calls made in your AWS environment.

What CloudTrail Captures:

- API calls from AWS Management Console
- AWS Command Line Interface (CLI) operations
- SDK and API requests
- Service-specific events

Event History: CloudTrail provides 90 days of event history by default without creating a trail. For longer retention, you must create a trail that delivers events to S3.



CloudTrail Key Features

Multi-Region Trails

A single trail can log events across all AWS regions, providing comprehensive visibility into global account activity.

Data Events

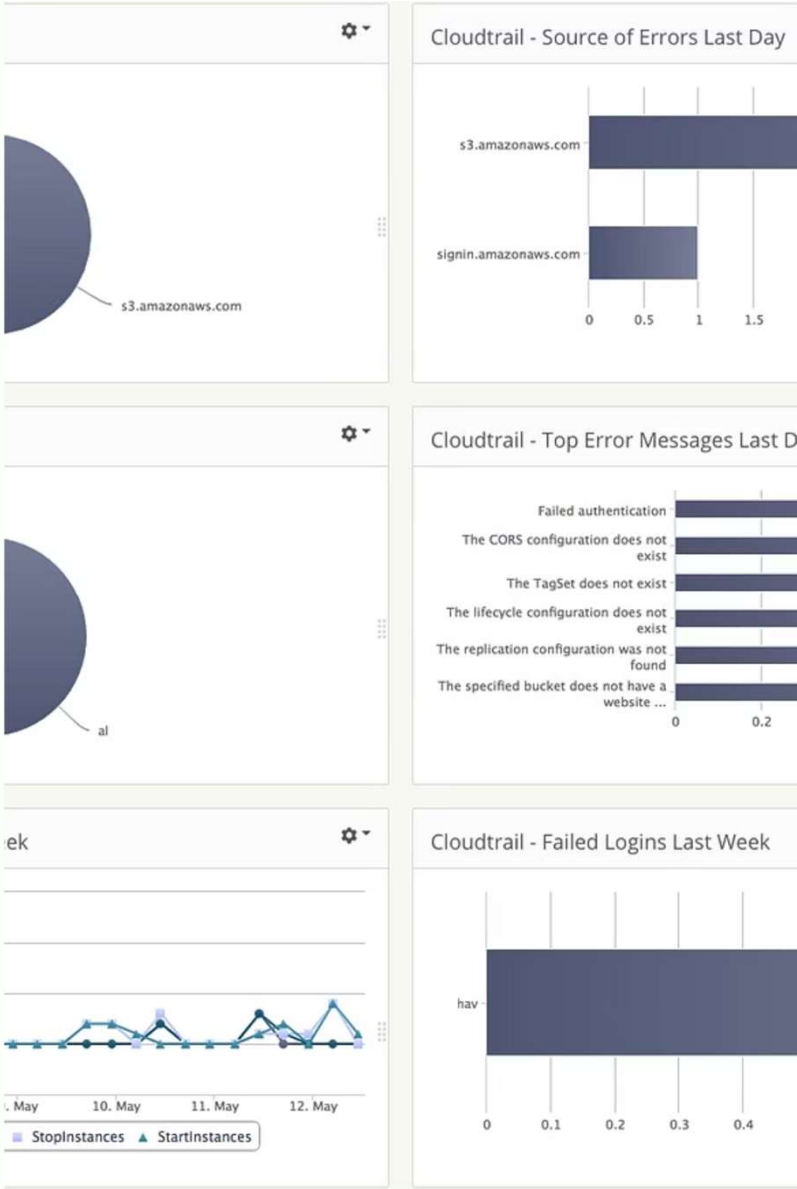
Track object-level operations on S3 buckets, Lambda function executions, DynamoDB tables, and more for detailed resource activity.

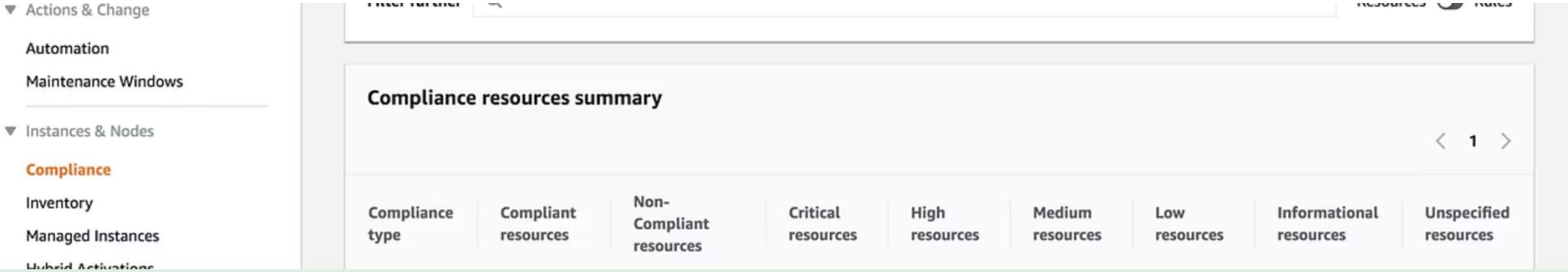
Service Integration

CloudTrail integrates with S3 for storage, CloudWatch Logs for monitoring, and EventBridge for automated responses to specific events.

Use Cases:

- **Security Forensics:** Investigate suspicious activity
- **Compliance Auditing:** Demonstrate regulatory compliance
- **Troubleshooting:** Identify configuration changes
- **Operational Analysis:** Track resource usage patterns





Part 3: AWS Config

Resource configuration tracking and compliance

AWS Config Overview

AWS Config provides a detailed view of the configuration of AWS resources in your account. It continuously monitors and records your AWS resource configurations and allows you to automate the evaluation of recorded configurations against desired configurations.

How AWS Config Works:

Record Configuration Changes

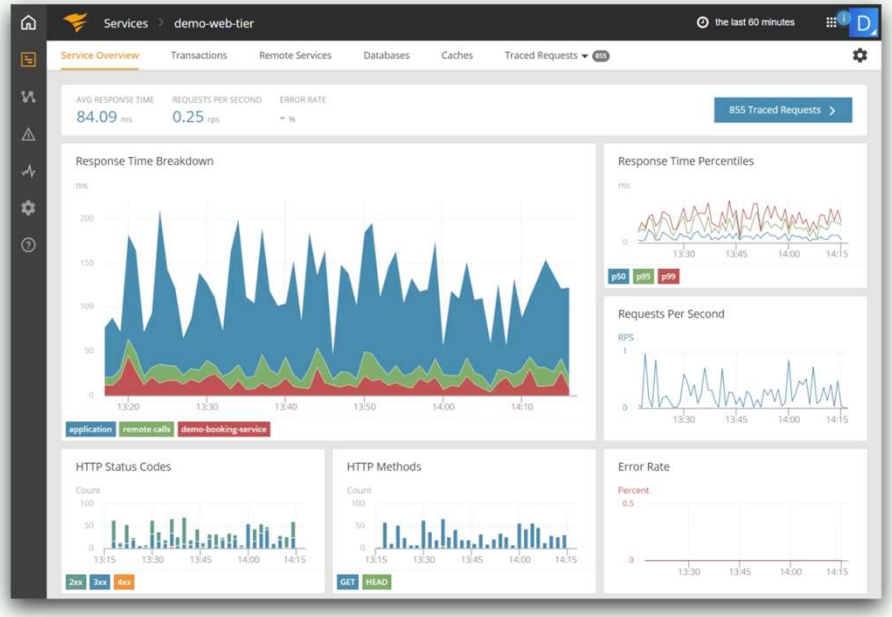
AWS Config continuously records changes to your resources as configuration items (CIs)

Evaluate Compliance

Config Rules check if resources comply with your desired configurations

Store Configuration History

Configuration snapshots and history are stored in an S3 bucket



AWS Config provides a timeline view of resource changes, making it easy to see what changed and when.

AWS Config Rules & Use Cases

Managed Rules Examples:

- **s3-bucket-public-read-prohibited:** Checks if S3 buckets allow public read access
- **required-tags:** Ensures resources have specific tags
- **ec2-instance-no-public-ip:** Checks if EC2 instances have public IPs
- **cloudtrail-enabled:** Verifies CloudTrail is enabled

Custom Rules: Create Lambda-based checks for organization-specific requirements

Key Use Cases:

Compliance Reports

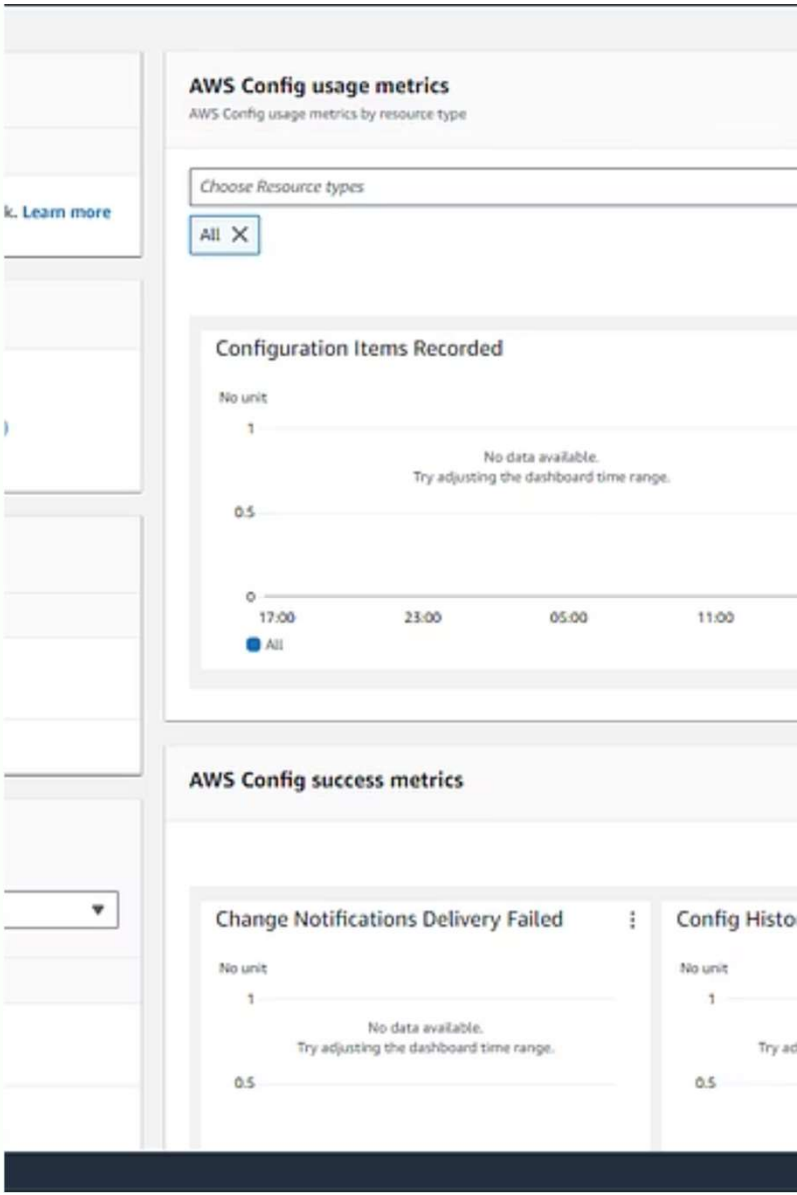
Generate reports for internal audits and regulatory requirements

Security Policy Enforcement

Ensure resources meet security standards

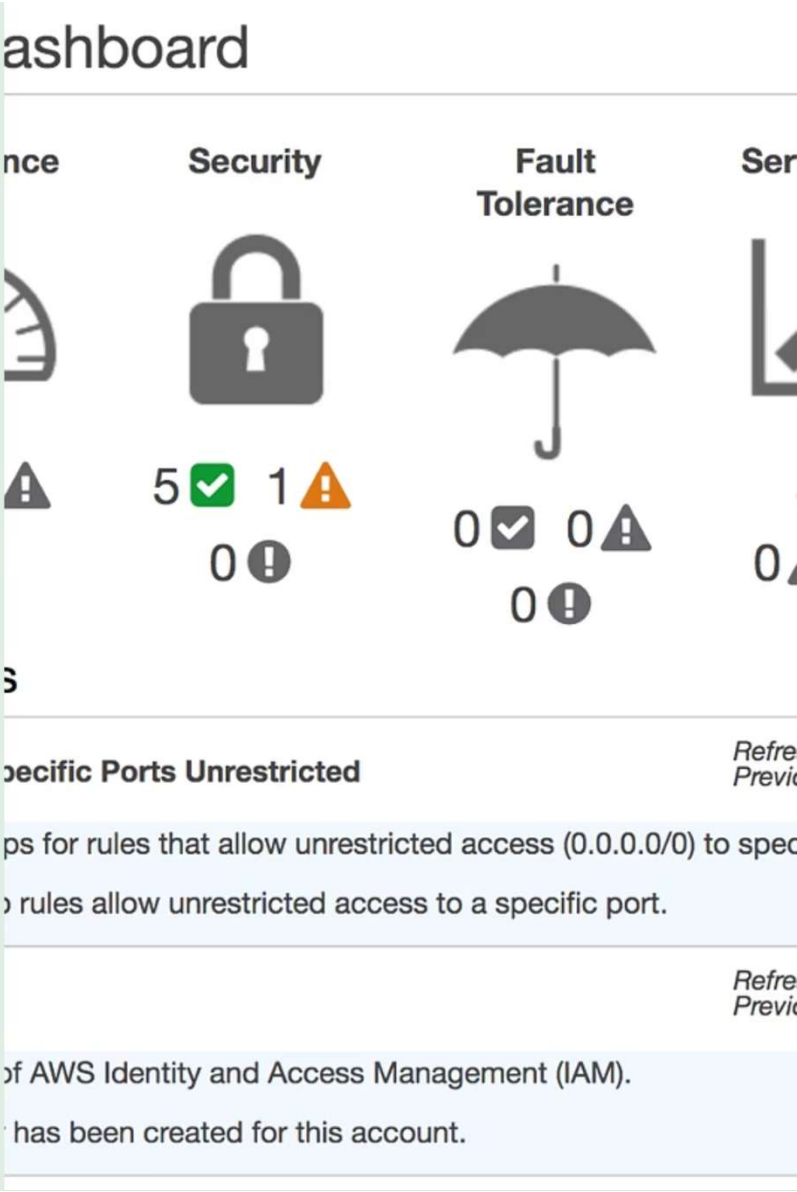
Change Tracking

Monitor and record configuration changes over time



Part 4: AWS Trusted Advisor

Best practice recommendations and optimization



Trusted Advisor Overview

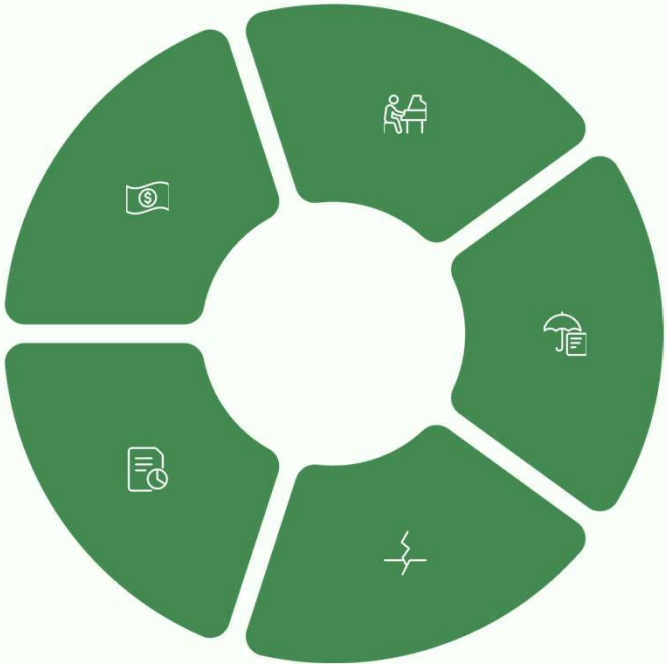
AWS Trusted Advisor is an online tool that provides real-time guidance to help you provision your resources following AWS best practices, improving performance, security, and reducing costs.

Cost Optimization

Identifies opportunities to eliminate unused and idle resources or make commitments to reserved capacity

Service Quotas

Checks for service usage that is more than 80% of the service quota



Performance

Helps improve the speed and responsiveness of your applications

Security

Identifies security configurations that could make your application vulnerable

Fault Tolerance

Helps improve your application's availability and redundancy

Trusted Advisor Example Checks

Security Checks

- MFA on root account
- IAM password policy
- Exposed access keys
- Security groups with unrestricted access

Cost Optimization

- Idle EC2 instances
- Underutilized EBS volumes
- Unassociated Elastic IP addresses
- RDS idle DB instances

Fault Tolerance

- S3 bucket versioning
- ELB connection draining
- Auto Scaling groups
- RDS Multi-AZ

Performance

- High utilization EC2 instances
- CloudFront alternate domain names
- Route 53 alias resource record sets
- EBS provisioned IOPS

Part 5: Summary & Q&A

Bringing it all together

Recap: AWS Monitoring & Logging Tools

CloudWatch

Monitor resources and applications with metrics, logs, and alarms

- Real-time monitoring
- Log aggregation
- Automated actions

CloudTrail

Track API activity and user actions

- Security auditing
- Compliance verification
- Operational troubleshooting

AWS Config

Track resource configurations and compliance

- Configuration history
- Compliance rules
- Change management

Trusted Advisor

Best practice recommendations

- Cost optimization
- Security improvements
- Performance enhancements

These tools work together to provide comprehensive visibility, security, and optimization for your AWS environment.